Prischa Bajeli

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I am a Third-year B.Tech student specializing in Artificial Intelligence & Data Science. I have two publications in IEEE ETCOM 2025, Springer LNNS 2025, and I have experience in developing full-fledged ML models, data pipelines, and applications that range from embedding-based facial recognition using Raspberry Pi to deployment of a FastAPI application that predicts Chest X-ray results. I am proficient in Python, PyTorch, Keras, SQL, scikit-learn.

SKILLS

Languages: Python, C, C++, SQL

ML & AI: PyTorch, Keras, TensorFlow, scikit-learn, LSTM, DenseNet121, Computer Vision, NLP

Data & Visualization: pandas, NumPy, Matplotlib, Excel, IBM Cognos

Tools & Platforms: FastAPI, Git, GitHub, Jupyter, Raspberry Pi

Soft Skills: Technical writing, Research & experimentation, Leadership, Communication

PROJECTS

IoT-Enabled Face Recognition Attendance System | Github Project Blog

Python | OpenCV | LBPH | Raspberry Pi 4 | GPIO | SQLite

- Implemented an LBPH face recognition system on Raspberry Pi 4 to automate attendance tracking for 50+ students, reducing end-to-end logging time from ~5 minutes per session to zero human input.
- Built end-to-end LBPH face recognition pipeline on Raspberry Pi 4 to achieve 95%+ accuracy on 50+ student profiles and eliminating all manual attendance entry for sessions of 50+ students.
- Research based on this work published at IEEE ETCOM 2025. Link

Multivariate Time-Series Energy Consumption Prediction | Github Project Blog

Python | Keras | TensorFlow | LSTM | scikit-learn | pandas | NumPy | Matplotlib

- RFE-LSTM forecasting pipeline developed using 35K hourly energy data samples; ablation study performed on the number of features and windows, resulting in a decrease of up to 7% in MAE/RMSE compared to the no-lag version, where lag features (lag 1–3) were found to be the important signals.
- Framed the forecasting/prediction task as a production-ready energy management tool. The ablation results were documented.
- Research based on this work published at ICTCS 2025. Link

Explainable AI for Chest X-Ray Diagnosis (Deep Learning + XAI) Github

Python | PyTorch | DenseNet121 | FastAPI | Grad-CAM | NumPy | Matplotlib

- Design of multi-label DenseNet121 classifier for NIH ChestX-ray14 dataset (112k images and 14 pathology labels) using stratification of patients for each split to avoid data leakage; obtained 0.83 macro-AUC without any pre-training tricks that match existing benchmark results.
- Threshold calibration for label-specific classifers by ROC curves on validation dataset for optimizing F1 score of rare pathology detection while minimizing the false negative rate in critical cases.
- Deployment of trained model via a FastAPI REST endpoint with a web UI allowing non-technical personnel to visualize predictions via Grad-CAM along with probability distribution of predicted labels.

JobAI — AI-Powered Career Copilot | Github Live App

Github | Live App TypeScript | React Native | Expo | Express.js | PostgreSQL | Drizzle ORM | Anthropic Claude API | TanStack React Query

- DenseNet121 multi-label classifier design for NIH ChestX-Ray14 database (containing 112k images and 14 pathologies) using patient stratification for each data split to avoid leakage; achieved 0.83 macro-AUC score without any tricks during pre-training and without dropping benchmark results.
- Label-specific classifiers threshold tuning by ROC curve on validation set aimed at optimizing F1 score for rare pathology recognition with minimal false negatives.
- Deployment of the trained model through a REST endpoint provided by FastAPI framework alongside a graphical user interface facilitating non-technical users in visualization of the prediction output using Grad-CAM technique together with prediction probability.

PUBLICATIONS

Prischa Bajeli et al., "IoT-Enabled Face Recognition Attendance System," IEEE ETCOM 2025. Link

Prischa Bajeli et al., "Multivariate Time-Series Energy Consumption Prediction," Springer LNNS, ICTCS 2025, Jaipur, India. Link

EDUCATION

B.Tech - Computer Science & Engineering (Artificial Intelligence & Data Science)

MIT World Peace University Pune, India • 2023 - Present

- **Current CGPA**: 8.20
- Relevant courses: Data structures and algorithms, DBMS, Data engineering, AI and Expert systems, Operating Systems, Machine learning, Theory of computation, Deep learning models, Cognitive computing and NLP processing, UI/UX

Class 12: 84% | **Class 10:** 94%

CERTIFICATIONS

Al Fundamentals IBM SkillsBuild • 2024

Data Visualizations and Building Dashboards with Excel and Cognos • 2024

NPTEL - Introduction to Large Language Models • 2026